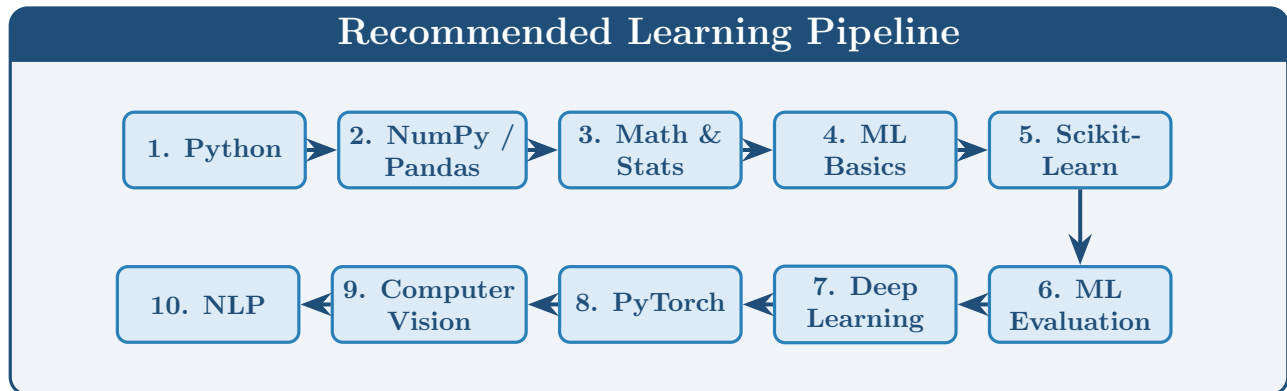


AI / ML Learning Roadmap

From Beginner Fundamentals to Natural Language Processing

1. Learning Dependency Flow



2. Detailed Curriculum & Modules

1. Python Programming

- ◇ **Basics:** Syntax, Variables, Data Types, Operators, Conditions, Loops, Functions
- ◇ **Data Structures:** Strings, Lists, Tuples, Sets, Dictionaries, List Comprehensions
- ◇ **Core Concepts:** Exception Handling, File Handling, Modules, Packages, `pip`, Virtual Envs
- ◇ **Advanced:** OOP, Iterators, Generators, Decorators, Lambda Functions, Type Hints

2. Python for Data Science

- ◇ **NumPy:** Arrays, Dimensions, Indexing, Slicing, Data Wrangling, Statistical Operations, Etc.
- ◇ **Pandas:** Series, DataFrames, Data I/O, Filtering, Grouping, Merging, Missing Values
- ◇ **Visualization & Analysis:** Matplotlib, Seaborn, Exploratory Data Analysis (EDA)

3. Mathematics & Statistics

- ◇ **Linear Algebra:** Vectors, Matrices, Operations, Transformations
- ◇ **Calculus:** Derivatives, Partial Derivatives, Gradients, Optimization
- ◇ **Probability & Stats:** Distributions, Sampling, Descriptive Stats, Hypothesis Testing

4. Machine Learning Fundamentals

- ◇ **Core Concepts:** Features, Labels, Supervised vs. Unsupervised Learning
- ◇ **Model Lifecycle:** Dataset Splitting (Train/Val/Test), Overfitting vs. Underfitting
- ◇ **Theory:** Bias-Variance Tradeoff, Feature Engineering, Feature Selection

5. Scikit-learn & ML Algorithms

- ◇ **Preprocessing:** Scaling, Encoding, Pipelines, Preprocessing Rules
- ◇ **Algorithms:** Linear/Logistic Regression, KNN, Naive Bayes, Decision Trees, SVM
- ◇ **Ensembles:** Random Forest, Bagging, Boosting (XGBoost, LightGBM, CatBoost)

6. ML Evaluation & Optimization

- ◇ **Metrics:** MAE, MSE, RMSE, R^2 , Precision, Recall, F1-Score, ROC-AUC
- ◇ **Tuning:** Cross-Validation, GridSearchCV, RandomizedSearchCV
- ◇ **Interpretability:** Feature Importance, Model Interpretability, SHAP Values

7. Deep Learning Fundamentals

- ◇ **Neural Networks:** Perceptrons, Layers, Activation Functions, Forward/Backprop
- ◇ **Training Mechanics:** Loss Functions, Gradient Descent, Optimizers, Learning Rates
- ◇ **Regularization:** Dropout, Batch Normalization, Epochs & Batches

8. PyTorch Framework

- ◇ **Core API:** Tensors, Autograd, `nn.Module`, Datasets & DataLoaders
- ◇ **Workflow:** Training/Validation Loops, GPU/CUDA Acceleration
- ◇ **Deployment:** Model Saving/Loading, Transfer Learning Practices

9. Computer Vision (CV)

- ◇ **Fundamentals:** Image Formats, OpenCV, Image Processing & Augmentation
- ◇ **Architectures:** CNNs, Object Detection (YOLO, IoU, NMS)
- ◇ **Tasks:** Image Classification, Semantic & Instance Segmentation

10. Natural Language Processing (NLP)

- ◇ **Text Preprocessing:** Tokenization, Stop Words, Stemming, Lemmatization
- ◇ **Vectorization:** Bag of Words, TF-IDF, Word Embeddings (Word2Vec)
- ◇ **Sequential Models:** RNNs, LSTMs, GRUs